The GSLV MARK III launcher aims to slowly phase out all the other space launchers being used

ious satellites in the geostationary orbit and will also be used to launch the first Indian crew vehicle in the future.

Satellite launches:

- **GSAT9** - The latest addition to the indigenously developed GSAT (Geosynchronous satellites) family of satellites, the GSAT 9 is a multi band observation and communication satellite slated for a launch before the end of this year. The 2.3 tonne satellite will carry two payloads, a 12 Ku band transponder, and the GAGAN system that will aims to provide GPS services to Indian Security forces and the Air traffic Control Organisation. The GAGAN (GPS Aided Augmented Navigation) system cost around ₹7.74 billion. And has been under development since 2008. A joint project being developed by Airport Authority of India and the Indian Space Research Organisation, the system will help pilots flying over the Indian airspace to navigate within an accuracy of 3 m. The system will also aid in the study of Ionosphere over the Indian region.
- **GSAT11** - Another of India's communication satellite, the GSAT-11 has a launch date set fot the year 2017. The 4.5 tonne satellite, weighing twice as much as the heaviest Indian satellite in orbit, will most likely be launched aboard the new GSLV-Mark III and carry a payload consisting of 40 high power Ku, Ka band transponders that are currently being built at the Space Application Center in Ahmedabad. The satellite's main aim is to provide advanced telecom and DTH services in India. Capable of transmitting data at a speed of 10 Gigabytes per second, the satellite will take care of the entire country's communication and broadcasting needs and is a big step for India.
- **NISAR** - A joint project by NASA and ISRO, the NASA-ISRO Synthetic Aperture Radar (NISAR) system aims to co-develop and launch a first of its kind radar imaging satellite that will use dial frequency. The payload will consist of two Synthetic Aperture Radars (SAR), the L-band SAR provided by NASA and the S band SAR made by ISRO. The mission aims to measure some of the most complex phenomenon happening on the earth's surface including natural hazards like earthquakes, volcanoes and tsunamis along with insights on the evolution and state of Earth's crust and information about the ongoing climate change crisis.

Other Space Exploration missions:

- **Chandrayan-2** - The successor to Chandrayaan-1, India's first lunar exploration mission which propelled India's space programme into international limelight, the Chandrayaan-2 is being developed by ISRO and is slated for a launch by the year 2018 using the newly developed heavy lift launcher GSLV Mark III. The 2nd lunar exploration mission, is going to include a lunar orbiter along with a a lander and rover. Initially, both the lander and rover for the mission were supposed to be developed by Russia, but when the Soviets stated their inability to provide them by 2015, the deadline of the agreed timeframe, Indian scientist took it upon themselves to develop them in-house. The design of the lander has been successfully completed bt the Space Application Centre in Ahmedabad while the Rover is being designed by ISRO at IIT kanpur facilities, making it a completely

Chandrayaan 2 aims to land a rover on moon this time

homegrown mission. The orbiter will be carrying a variety of payloads for which it will include a number of scientific instruments. The main aim of the mission is to develop advanced lander and rover technologies and set the basis for future projects such as the Mangalyaan-2 which will see India landing a Rover/lander on Mars.
- **Mangalyaan 2** - India's much talked about Mars Orbiter Mission was probably one of its most successful ones helping it set a number of records including making India the 'first Asian nation to reach Mars orbit", "the fourth nation in the world to reach Marsorbit", "first nation in the world to do so in its first attempt" and the most important of all, "the least expensive Marsmission to date". Mangalyaan 2 is the successor mission which aims to land a rover and lander on Mars alongside